NT TECH INNOVATION

FULL STACK DEVELOPER | PRACTICAL ASSESSMENT

Assessment: Independent Full-Stack Project **Submission Deadline:** 19 October 2026

1. Objective

Choose a practical software idea that solves a real problem and build a working full-stack application. Your project may address any industry or business use case. We will assess your ability to plan, implement, explain, and improve your solution independently.

2. Required Implementation

• **Frontend:** Responsive, usable interface with clear loading, success, and error states.

• **Backend and data:** Working APIs, database persistence, meaningful CRUD operations, and appropriate relationships.

• **Authentication and security:** Secure login and protected operations; apply authorization and input validation where relevant.

• **Business logic:** Implement at least one meaningful workflow, calculation, or status-based process.

• **AI / ML feature:** Include one practical AI-powered or basic machine-learning feature relevant to your project (for example, smart recommendations, classification, text summarization, or demand prediction). An existing AI API, open-source model, or a simple self-built model may be used. Explain the inputs, outputs, limitations, and how you evaluated its usefulness.

3. Technical Expectations

Use a suitable technology stack of your choice. Keep code modular, use Git with meaningful commits, handle invalid inputs, and include basic testing of a critical feature. Prioritize a reliable working application over excessive scope.

4. Submission and Live Review

• Submit a Git repository link, README with setup instructions, environment-variable example, and demo credentials if required.

• Provide a working deployment link if available; otherwise be ready to run the application locally.

• Demonstrate the main user flow and the AI/ML feature. Explain the database, key APIs, architecture, and known limitations.

• Be prepared for a short live code change, debugging task, and technical questions about your implementation.

5. Evaluation (100 Marks)

Problem Solving & Planning	10	Security & Validation	10
Frontend & User Experience	15	AI / ML Feature & Explanation	10
Backend, APIs & Business Logic	20	Code Quality, Testing & Git	10
Database & Data Handling	15	Live Demonstration & Technical Review	10

6. Independence Policy

The solution and submitted code must be your own work. AI coding assistants, generated project code, and copied complete solutions are not permitted. Using an AI API or ML model **as part of your application's functionality** is allowed. You must be able to explain and modify every submitted component.